National University of Computer and Emerging Sciences
Islamabad Campus

0058

Web Programming (CS4032)

Sessional-I Exam

Course Instructor(s):
Dr. Tajwar Mehmood
Section(s): (if applicable) A, B, C
Spring 26

Total Time (Hrs): 1
Total Marks: ~~80~~ 52
Total Questions: 6

Date: Feb 21, 2026

23I-0575 — **Roll No**

G(A) — **Course Section**

Student Signature

Do not write below this line.

Attempt all the questions.

CLO 1. Explain the role and significance of modern web application development technologies, tools, and industry best practices.

CLO 2. Design and develop dynamic, responsive, and scalable web applications using modern front-end and back-end frameworks.

NOTE: Complete the missing parts (_____) to create a simple asked task.

[CLO 2]

Q1: Student Grade Calculator JavaScript Code. [10 marks]

1. Takes a student's name
2. Takes marks for 3 subject
3. Calculates the average
4. Displays the grade

Fill in the Blanks

10

Code:

```
<h2>Student Grade Calculator</h2>
Name: <input type="text" id="studentName"><br><br>
Math: <input type="number" id="math"><br><br>
Science: <input type="number" id="science"><br><br>
English: <input type="number" id="english"><br><br>
<button onclick="calculateGrade()">Calculate</button>
<h3 id="result"></h3>
<script>
  function calculateGrade() {
    let name = document.getElementbyId("studentName").value;
    let math = Number(document.getElementbyId("math").value;
    let science = Number(document.getElementbyId("science").value;
    let english = Number(document.getElementbyId("english").value;
    let average = (math+science+english)/3;
    let grade;
```


10 + 10 + 2 + 10 + 10 + 10

```
        if (average >= __90__) {
            grade = "A";
        } else if (average >= __80__) {
            grade = "B";
        } else if (average >= __70__) {
            grade = "C";
        } else {
            grade = "Fail";
        }

        document.getElementById("__result__").innerHTML ="Student: " + name +
            "<br>Average: " + average.toFixed(2) +
            "<br>Grade: " + grade;
    }
</script>
```

[CLO 2]

Q2: Smile Toggle Task using Jquery . Complete the blanks to create a small web page where: [10 ma

- Clicking the button changes 😐 to ☺
- Clicking again changes it back
- Uses .text() and .click()

```
<style>
body {
text-align: center;
font-family: Arial;
margin-top: 100px;
}

#face {
font-size: 80px;
}

button {
padding: 10px 20px;
font-size: 16px;
}
</style>
</head>
<body>

<h2>Smile Toggle</h2>

<div id="face">😐</div>
<br>
<button id="smileBtn">Change Mood</button>
```

```
$(document).ready(function() {
$("#smileBtn").click(function() {

let currentFace = $("#face").text();

if (currentFace === "😐") {
$("#face").text("☺");
} else {
$("#face").text("😐");
} }); });
</script>
```

10

[CLO 1]

Q3: Write the output of following script in Asynchronous flow **[2 marks]**

```
<script>
function myDisplayer(some) {
  document.getElementById("demo").innerHTML += some + " ";
}
myDisplayer("A");
setTimeout(function() {
  myDisplayer("B");
}, 1000);
myDisplayer("C");
</script>
```

Answer:

Some A, Some C Some B

2

[CLO 2]

Q4:Match the following images with their code snippet

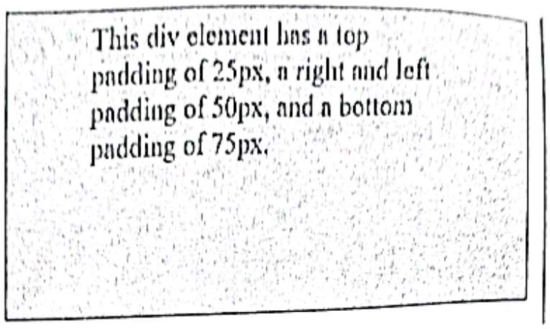

Image 1

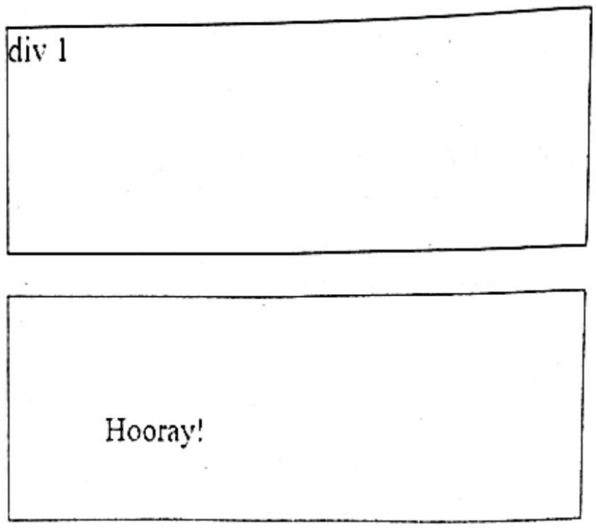

Image 2

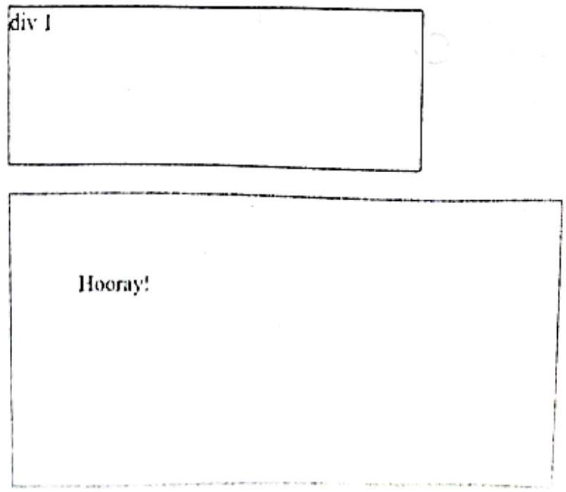

Image 3

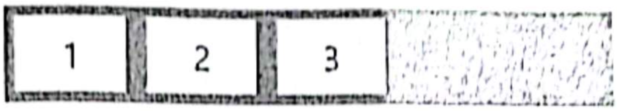

Image 4

Image 5

Code Snippet	Image Number

<style>
div {
  border: 1px solid black;
  padding: 5px 50px 75px;
  background-color: lightblue;
}
</style>
	1

<style>
.div1 {
  width: 300px;
  height: 100px;
  border: 1px solid blue;
  box-sizing: border-box;
}
.div2 {
  width: 300px;
  height: 100px;
  padding: 50px;
  border: 1px solid red;
  box-sizing: border-box;
} </style> </head>
<body>
<div class="div1">div 1</div>
 	
<div class="div2">Hooray!</div></body>
	2

<style>
.div1 {
  width: 300px;
  height: 100px;
  border: 1px solid blue;	

}
.div2 {
  width: 300px;
  height: 100px;
  padding: 50px;
  border: 1px solid red;
}
</style>
</head>
<body>
<div class="div1">div 1</div>
 	
<div class="div2">Hooray!</div>
</body>
	3

.flex-container {
  display: flex;
  flex-direction: row;
}
	4

<style>
#example1 {
  background-color: lightblue;
  height: 100px;
  width: 200px;
  border-start-end-radius: 50px;
}
</style>
	5

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[CLO 1]

Q5: Write the code for Resize the browser window to see the effect. **[10 marks]**

On screens that are 992px wide or less, the columns will resize from four columns to two columns. On screens that are 600px wide or less, the columns will stack on top of each other instead of next to each other.

Complete the blanks:

```
<head>
<meta name="Viewport" content="width=device-width, initial-scale=1">
<style>
* {
 box-sizing: border-box;
}

/* Container for flexboxes */
```

2

0

```
flex ;
p: wrap;

create four equal columns */
.column {
flex: 25 %;
padding: 20px;
}

/* On screens that are 992px wide or less, go from four columns to two columns */
@media screen and (max-width: 992px) {
  .column {
    flex: 50 %;
  }
}

/* On screens that are 600px wide or less, make the columns stack on top of each other instead of next to each other */
@media screen and (max-width: 600px) {
  .container {
    flex-direction: stack ;
  }
}
</style>
</head>
<body>
<div class="container">
  <div class="column" style="background-color:#aaa;">
    <h2>Column 1</h2>
    <p>Some text..</p>
  </div>

  <div class="column" style="background-color:#bbb;">
    <h2>Column 2</h2>
    <p>Some text..</p>
  </div>

  <div class="column" style="background-color:#ccc;">
    <h2>Column 3</h2>
    <p>Some text..</p>
  </div>

  <div class="column" style="background-color:#ddd;">
    <h2>Column 4</h2>
    <p>Some text..</p>
  </div>
</div>

</body>
```

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[CLO 1]

Q6: Write the following code using same flow with async and await [10 marks]

```
<body>
<h1>JavaScript Promises</h1>
<h2>Running Functions in Steps</h2>

<p id="demo"></p>
<script>
// Function to display something
function myDisplayer(some) {
  document.getElementById("demo").innerHTML = some;
}
// Three functions to run in steps
function step1() {
  return Promise.resolve("A");
}
function step2(value) {
  return Promise.resolve(value + "B");
}
function step3(value) {
  return Promise.resolve(value + "C");
}
// Run the three functions in steps
step1()
.then(function(value) {
  return step2(value);
})
.then(function(value) {
  return step3(value);
})
.then(function(value) {
  myDisplayer(value);
});
</script>
]
```

Answer :

```
<script>
  function myDisplayer(some) {
      document.getElement byId ("demo").innerHTML=some
  }
  async function step1 () {
      return "A";
  }
async function step2(...
{
    return Valu...
}
async function step3(...
{
  return Value + "C";
}
try {
 let a = await step1();
let b = await step2(a);
let c = await step3(b);
myDisplayer(c);
}
catch (err) { }

 </script>
```

not necessary ⟹

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